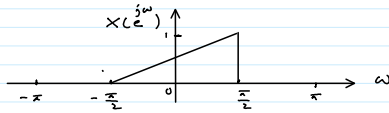


Example: The DTFT of $x[n]$ is shown below:



Let, $y(t) = \sum_{n=-\infty}^{+\infty} x[n] e^{j2\pi n t}$

Find, $I = \int_0^{\pi/2} y(t) dt$

Solution: $y(t) = \sum_{n=-\infty}^{+\infty} x[n] e^{j2\pi n t} \xrightarrow{m=-n} y(t) = \sum_{m=-\infty}^{+\infty} x[m] e^{-j2\pi m t} \quad (I)$

Knowing: $X(e^{j\omega}) = \sum_{m=-\infty}^{+\infty} x[m] e^{-j\omega m} \quad (II)$

then $y(t) = X(e^{j2t})$

therefore: $I = \int_0^{\pi/2} y(t) dt = \int_0^{\pi/2} X(e^{j2t}) dt$

Let $t' = 2t$

$I = \int_0^{\pi} X(e^{jt'}) \frac{dt'}{2} = \frac{1}{2} \int_0^{\pi} X(e^{jt'}) dt' \quad (1)$

The area of $X(e^{jt'})$ in $0 < t' < \pi$ is



$S = \int_0^{\pi/2} X(e^{jt'}) dt' = \frac{1}{2} \left(\frac{\pi}{2} \right) \left(1 + \frac{1}{2} \right) = \frac{3\pi}{8}$

Finally, using Eq. (1): $I = \frac{S}{2} = \frac{3\pi}{16}$

example: Let $x[n] \longleftrightarrow X(e^{j\omega})$ and is given by

$X(e^{j\omega}) = \sum_{k=-\infty}^{+\infty} \frac{\sin\left(\frac{3k\pi}{4}\right)}{\sin\left(\frac{k\pi}{4}\right)} \delta\left(\omega - \frac{k\pi}{2}\right)$

Find $x[5]$?

Solution: method 1:

Use Inverse DTFT of $X(e^{j\omega})$

$x[n] = \frac{1}{2\pi} \int_0^{2\pi} X(e^{j\omega}) e^{j\omega n} d\omega$

in $0 \leq \omega \leq 2\pi$ $X(e^{j\omega})$ is

$X(e^{j\omega}) = \sum_{k=0}^3 \frac{\sin\left(\frac{3k\pi}{4}\right)}{\sin\left(\frac{k\pi}{4}\right)} \delta\left(\omega - \frac{k\pi}{2}\right)$

$= \underset{k=0}{3\delta(\omega)} + \frac{\sin\left(\frac{3\pi}{4}\right)}{\sin\left(\frac{\pi}{4}\right)} \delta\left(\omega - \frac{\pi}{2}\right) + \frac{\sin\left(\frac{3\pi}{2}\right)}{\sin\left(\frac{\pi}{2}\right)} \delta\left(\omega - \pi\right) + \frac{\sin\left(\frac{9\pi}{4}\right)}{\sin\left(\frac{3\pi}{4}\right)} \delta\left(\omega - \frac{3\pi}{2}\right)$

$= 3\delta(\omega) + \delta\left(\omega - \frac{\pi}{2}\right) - \delta\left(\omega - \pi\right) + \delta\left(\omega - \frac{3\pi}{2}\right) \quad 0 \leq \omega \leq 2\pi$

$x[5] = \frac{1}{2\pi} \int_0^{2\pi} X(e^{j\omega}) e^{j5\omega} d\omega$

$= \frac{1}{2\pi} \int_0^{2\pi} \left[3\delta(\omega) + \delta\left(\omega - \frac{\pi}{2}\right) - \delta\left(\omega - \pi\right) + \delta\left(\omega - \frac{3\pi}{2}\right) \right] e^{j5\omega} d\omega$

$= \frac{1}{2\pi} \left(3 + e^{j5\frac{\pi}{2}} - e^{j5\pi} + e^{j5\frac{3\pi}{2}} \right)$

$= \frac{1}{2\pi} (3 + j + 1 - j) = \frac{2}{\pi}$

method #2

From $X(e^{j\omega}) \in \mathcal{E}_8$:

$$X(e^{j\omega}) = \sum_{k=0}^3 \frac{\sin(\frac{3k\pi}{4})}{\sin(\frac{k\pi}{4})} \delta(\omega - \frac{k\pi}{2})$$

We know $x[n]$ is periodic $\cancel{K} \frac{2\pi}{N} = \cancel{K} \frac{\pi}{2} \Rightarrow N=4$
 $\omega_0 = \frac{2\pi}{N} = \frac{2\pi}{4} = \frac{\pi}{2}$

Compare with:

$$X(e^{j\omega}) = \sum_{k \in \langle N \rangle} 2\pi a_k \delta(\omega - k\omega_0)$$

$$\therefore \omega_0 = \frac{\pi}{2} \quad \& \quad a_k = \frac{1}{2\pi} \frac{\sin(\frac{3k\pi}{4})}{\sin(\frac{k\pi}{4})}$$

Recall:

$$y[n] \xleftrightarrow{\text{F.S.}} b_k = \frac{A}{N} \frac{\sin\left[\left(\frac{2\pi}{N}\right)k\left(N_1 + \frac{1}{2}\right)\right]}{\sin\left(\frac{\pi}{N}k\right)} = \frac{A}{N} \frac{\sin\left(\frac{(2N_1+1)k\pi}{N}\right)}{\sin\left(\frac{k\pi}{N}\right)}$$

in this problem

$$a_k = \frac{1}{2\pi} \frac{\sin(\frac{3k\pi}{4})}{\sin(\frac{k\pi}{4})}$$

$x[n]$ is periodic with $N=4$

$$a_k = \frac{1}{2\pi} \frac{\sin(\frac{3k\pi}{4})}{\sin(\frac{k\pi}{4})} = \frac{A}{N} \frac{\sin\left(\frac{(2N_1+1)k\pi}{N}\right)}{\sin(\frac{k\pi}{N})}$$

$$N=4$$

$$= \frac{A}{4} \frac{\sin\left(\frac{(2N_1+1)k\pi}{4}\right)}{\sin(\frac{k\pi}{4})}$$

$$\Rightarrow \frac{A}{4} = \frac{1}{2\pi} \Rightarrow N_1=1 \quad A = \frac{2}{\pi}$$

$\therefore$



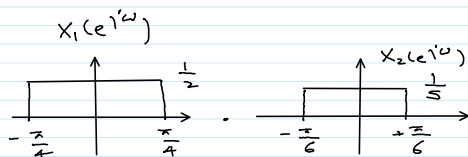
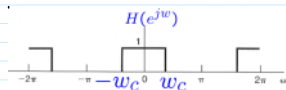
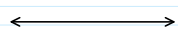
$$\therefore x[5] = \frac{2}{\pi}$$

Example: Parseval theorem:

$$\text{Find the summation: } A = \sum_{n=-\infty}^{+\infty} \frac{\sin(\frac{\pi n}{4})}{2\pi n} \cdot \frac{\sin(\frac{\pi n}{8})}{5\pi n}$$

Solution: Recall:

$$h[n] = \frac{\sin \omega_c n}{\pi n}$$



Parseval's relationship:

$$A = \sum_{n=-\infty}^{+\infty} x_1[n] \cdot x_2[n]^* = \frac{1}{2\pi} \int_{-\pi}^{+\pi} X_1(e^{j\omega}) X_2^*(e^{j\omega}) d\omega$$

$$A = \sum_{n=-\infty}^{\infty} x_1[n] \cdot x_2[n] = \frac{1}{2\pi} \int_{-\pi}^{\pi} X_1(e^{j\omega}) X_2(e^{j\omega}) d\omega$$

$$= \frac{1}{2\pi} \int_{-\frac{\pi}{6}}^{+\frac{\pi}{6}} \left(\frac{1}{2}\right) \left(\frac{1}{5}\right) d\omega = \frac{1}{60}$$

22 **Example:** The DTFT of $x[n]$ is given below;

$$\text{DTFT}\{x[n]\} \equiv X(e^{j\omega}) = \sum_{k=-\infty}^{+\infty} (-1)^k \delta\left(\omega - \frac{k\pi}{4}\right)$$

Find $x[4]$?

Solution: In the interval of $0 \leq \omega \leq 2\pi$ we have:

$$X(e^{j\omega}) = \sum_{k=0}^7 (-1)^k \delta\left(\omega - \frac{k\pi}{4}\right) \quad \text{for } 0 \leq \omega \leq 2\pi$$

8 impulses

$$x[4] = \frac{1}{2\pi} \int_0^{2\pi} X(e^{j\omega}) e^{j4\omega} d\omega$$

$$= \frac{1}{2\pi} \int_0^{2\pi} \left[\sum_{k=0}^7 (-1)^k \delta\left(\omega - \frac{k\pi}{4}\right) \right] e^{j4\omega} d\omega$$

$$= \frac{1}{2\pi} \sum_{k=0}^7 (-1)^k \int_0^{2\pi} \delta\left(\omega - \frac{k\pi}{4}\right) e^{j4\omega} d\omega$$

$$= \frac{1}{2\pi} \sum_{k=0}^7 (-1)^k e^{jk\pi} = \frac{1}{2\pi} \sum_{k=0}^7 (-1)^{2k} = \frac{1}{2\pi} (8) = \frac{4}{\pi}$$

25 **Example:** The impulse response ($h[n]$) of a D.T. LTI System is given by the following,

$$h[n] = \begin{cases} (-\alpha)^{-n} & n \leq 0 \\ 0 & n > 0 \end{cases}$$

Find the difference Equation of this system.

Solution: $h[n]$ can be written by $\Rightarrow g[n] = h[-n] = \begin{cases} (-\alpha)^n & n \geq 0 \\ 0 & n < 0 \end{cases}$

$$g[n] = (-\alpha)^n u[n] \xrightarrow{\text{DTFT}} G(e^{j\omega}) = \frac{1}{1 + \alpha e^{-j\omega}}$$

$$h[n] = g[-n] \Rightarrow H(e^{j\omega}) = G(e^{-j\omega}) = \frac{1}{1 + \alpha e^{j\omega}}$$

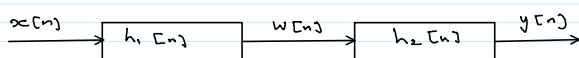
by Convolution property:

$$H(e^{j\omega}) = \frac{Y(e^{j\omega})}{X(e^{j\omega})} = \frac{1}{1 + \alpha e^{j\omega}}$$

$$(1 + \alpha e^{-j\omega}) Y(e^{j\omega}) = X(e^{j\omega})$$

$$y[n] + \alpha y[n+1] = x[n] \quad \leftarrow \text{Difference Eq.}$$

example: Consider the system given below



The frequency Response of first system is give by

$$H_1(e^{j\omega}) = e^{-j\omega} \cdot \begin{cases} 0 & ; \quad |\omega| \leq \frac{\pi}{4} \\ 1 & , \quad \frac{\pi}{4} < |\omega| \leq \pi \end{cases}$$

and the impulse response of the second system is shown by,

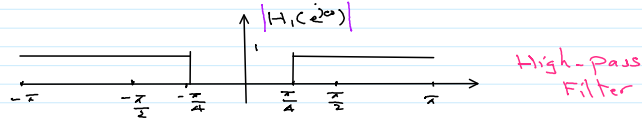
$$h_2[n] = 3 \frac{\sin\left(\frac{\pi}{2}(n-2)\right)}{\pi(n-2)}$$

What kind of filter is this system.

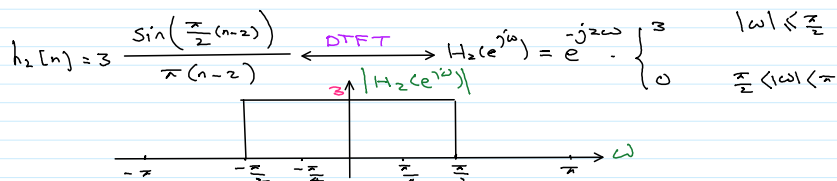
Solution:

notice:

$$h_{\text{overall}}[n] = h_1[n] * h_2[n] \Rightarrow H_{\text{overall}}(e^{j\omega}) = H_1(e^{j\omega}) \cdot H_2(e^{j\omega})$$



High-pass Filter



$$\begin{aligned} H_{\text{overall}}(e^{j\omega}) &= H_1(e^{j\omega}) \cdot H_2(e^{j\omega}) \\ &= e^{-j\omega} \cdot e^{-j2\omega} \cdot \begin{cases} 3 & , \quad \frac{\pi}{4} \leq |\omega| \leq \frac{\pi}{2} \\ 0 & \text{o/w} \end{cases} \\ &= e^{-j3\omega} \cdot \begin{cases} 3 & , \quad \frac{\pi}{4} \leq |\omega| \leq \frac{\pi}{2} \\ 0 & \text{o/w} \end{cases} \end{aligned}$$

∴ $H(e^{j\omega})$ is a Pass-band Filter

2nd

example Consider a System with the following eq.

$$\sum_{k=-\infty}^{+\infty} \left(\frac{1}{2}\right)^{|n-k|} y[k] = x[n]$$

What kind of system is this. Find the Freq. Response of this system

Solution:

$$\sum_{k=-\infty}^{+\infty} \left(\frac{1}{2}\right)^{|n-k|} y[k] = \left(\frac{1}{2}\right)^{|n|} * y[n] = x[n]$$

$$\frac{1 - \frac{1}{4}}{1 - \cos(\omega) + \frac{1}{4}} Y(e^{j\omega}) = X(e^{j\omega})$$

$$Y(e^{j\omega}) = \frac{5 - 4\cos(\omega)}{3} X(e^{j\omega})$$

$$H(e^{j\omega}) = \frac{1}{3} (5 - 4\cos(\omega)) = \frac{5}{3} - \frac{2}{3} e^{j\omega} - \frac{2}{3} e^{-j\omega}$$

$$h[n] = \frac{5}{3} \delta[n] - \frac{2}{3} \delta[n+1] - \frac{2}{3} \delta[n-1]$$

3rd

Example System LTI with its difference Eq. is represented by

$$y[n] = x[n] + 2x[n-1] - x[n-2]$$

Find the impulse response of a second system which is related to the first system by;

$$H_2(e^{j\omega}) = H_1(e^{j(\omega+\pi)})$$

Solution:

$$H_1(e^{j\omega}) = \frac{Y(e^{j\omega})}{X(e^{j\omega})} = (1 + 2e^{-j\omega} - e^{-j2\omega})$$

$$h_1[n] = \delta[n] + 2\delta[n-1] - \delta[n-2]$$

$$\begin{aligned} \therefore h_2[n] &= (-1)^n h_1[n] \\ &= (-1)^n (\delta[n] + 2\delta[n-1] - \delta[n-2]) \\ &= \delta[n] - 2\delta[n-1] + \delta[n-2] \end{aligned}$$

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example: Frequency Response of a LTI system is given by

$$H(e^{j\omega}) = \frac{1 + e^{-j2\omega}}{1 - 0.8e^{-j\omega}}$$

when input $x[n]$ is $x[n] = 4 + 3\cos(\omega_0 n)$

the output of this system is $y[n] = A$, (A is constant)

Find ω_0 .

Solution: Since $y[n] = A$, it indicates that the $\cos(\cdot)$ is filtered-out.

$\therefore$ the Frequency Response of this system at ω_0 should be

$$H(e^{j\omega}) \big|_{\omega=\omega_0} = 0$$

$$H(e^{j\omega_0}) = 0 \Rightarrow \frac{1 + e^{-j2\omega_0}}{1 - 0.8e^{-j\omega_0}} = 0 \Rightarrow 1 + e^{-j2\omega_0} = 0$$

$$e^{-j2\omega_0} = -1 = e^{-j\pi} \Rightarrow \omega_0 = \frac{\pi}{2}$$

Find DTFT of $x[n]$;

$$x[n] = [a^n \sin(\Omega_0 n)] u[n] \quad |a| < 1$$

Solution:
$$x[n] = a^n \left(\frac{e^{j\Omega_0 n} - e^{-j\Omega_0 n}}{2j} \right) u[n]$$

$$\begin{aligned} X(e^{j\omega}) &= \frac{1}{2j} \sum_{n=0}^{\infty} \left(a^n e^{jn\Omega_0} - a^n e^{-jn\Omega_0} \right) e^{-jn\omega} \\ &= \frac{1}{2j} \left[\frac{1}{1 - a e^{-j(\omega - \Omega_0)}} - \frac{1}{1 - a e^{-j(\omega + \Omega_0)}} \right] \end{aligned}$$

Finally:

$$x[n] \sin(\Omega_0 n) \longleftrightarrow \frac{j}{2} \left[X(e^{j(\omega + \Omega_0)}) - X(e^{j(\omega - \Omega_0)}) \right]$$

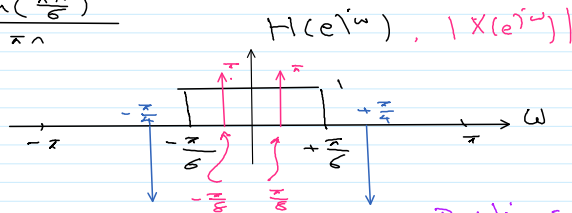
$$x[n] \sin(\Omega_0 n) \longleftrightarrow \frac{j}{2} \left[X(e^{j(\omega + \Omega_0)}) - X(e^{j(\omega - \Omega_0)}) \right]$$

$$x[n] \cos(\Omega_0 n) \longleftrightarrow \frac{1}{2} \left[X(e^{j(\omega + \Omega_0)}) + X(e^{j(\omega - \Omega_0)}) \right]$$

example: LTI system: Find output when $x[n]$ is given by

$$x[n] = \underbrace{\sin\left(\frac{\pi n}{8}\right)}_{N=6} - 2 \underbrace{\cos\left(\frac{\pi n}{4}\right)}_{N=8}$$

i) $h[n] = \frac{\sin\left(\frac{\pi n}{6}\right)}{\pi n}$



$$Y(e^{j\omega}) = X(e^{j\omega}) \cdot H(e^{j\omega})$$

$$y[n] = \sin\left(\frac{\pi n}{8}\right)$$

Periodic signals:

$$\sum_{k=-\infty}^{+\infty} 2\pi a_k \delta(\omega - k \frac{2\pi}{N})$$

ii) $h[n] = \frac{\sin\left(\frac{\pi n}{6}\right)}{\pi n} + \frac{\sin\left(\frac{\pi n}{2}\right)}{\pi n}$

then:

$$y[n] = 2 \sin\left(\frac{\pi n}{8}\right) - 2 \cos\left(\frac{\pi n}{4}\right)$$